Adijat Sulaimon

Embedded Systems & ML Engineer

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Authorised to work in Germany

Summary

Embedded Systems & ML Engineer with research experience in computer vision, deep learning and active learning, alongside project experience in real-time embedded systems, distributed software and formal verification. Experienced with Python, PyTorch, C/C++, FreeRTOS and UPPAAL.

Skills

Programming: Python, C, C++, VHDL
Machine Learning & Computer Vision: PyTorch, TensorFlow, scikit-learn, OpenCV, active learning, data augmentation
Embedded Systems & Real-Time: FreeRTOS, ESP32, Arduino, TCP/IP sockets, RMS scheduling, CUDA
Verification & Simulation: UPPAAL, UML/SysML, ModelSim, CARLA
Development Tools: Git, Docker, TensorBoard, Weights & Biases, SolidWorks

Experience

Research Assistant – ML/AI Nov 2023 – Present
Hochschule Hamm-Lippstadt (HSHL), Lippstadt, Germany

- Developed and evaluated ResNet, DenseNet, MobileNet, ViT and CLIP models in PyTorch for computer vision on imbalanced real-world datasets.
- Built an uncertainty-based active learning pipeline that reduced manual annotation effort by approximately 50% while maintaining model performance.
- Developed benchmarking and misclassification-analysis tools (Top-1/Top-5, per-class metrics); coordinated communication between ML and embedded workstreams.

Bachelor Thesis – Smart Labeling of Sugar Beet Plants Using Active Learning Jun 2023 – Oct 2023
HSHL, Lippstadt, Germany

- Extended a ResNet50-based uncertainty-sampling pipeline from binary to three-class classification using batch-mode sampling in PyTorch.
- Reached about 99% accuracy (3,076 of 3,102 images) on a separate test set, starting from roughly 10% labelled data per class.

Working Student – Interaction Design Feb 2022 – Mar 2023
Perfect-iD GmbH, Cologne, Germany

- Developed and implemented a design system that streamlined the design process and improved consistency across digital touchpoints, resulting in a 25% increase in conversions.
- Worked with the product and development teams in an agile startup environment to ensure designs were technically feasible and met business objectives.

QA/QC Engineer 2015 – 2019
RejiRoma Nigeria Ltd., Lagos, Nigeria

- QA on oil & gas projects (gas tank, pipeline, vapour recovery skid with PLC control): P&ID inspection, ISO 9001 documentation.

Projects

ADAS Perception & Emergency Braking in CARLA Sep 2024 – Feb 2025
CARLA, PyTorch, ResNet18/50, UPPAAL, Python | Group project (3)

- Integrated a ResNet-based front-camera classifier into a CARLA driver-assistance agent and evaluated its behaviour in closed-loop driving simulations.
- Trained ResNet18 and ResNet50 on about 2,000 images (85% and 81%); chose ResNet18 because ResNet50 made the simulation lag.
- Analysed failure cases, including pedestrian misclassification and late braking, identifying inference latency and dataset quality as potential contributors; first author of the project paper. Modelled left-turn decision logic in UPPAAL and checked five properties, including deadlock freedom.

Truck Platooning System 2024 – 2025

C++, TCP/IP, CUDA, Arduino, Tinkercad | Group project (4)

- Team built a leader–follower platoon in C++ with socket communication over TCP/IP, three threads per node and matrix clocks; seven unit tests passed.
- Performed the Rate Monotonic schedulability analysis (total utilisation 0.733 against the 0.779 bound for three tasks) and implemented a CUDA kernel for the inter-truck distance check.
- Wrote the requirement and parametric constraint diagrams (20 m minimum safe distance); simulated sensing on an Arduino Uno.

Edge AI: Split Neural Network Inference 2026

MobileNetV2, Python | M.Eng. project

- Evaluated 20 candidate split points for distributing MobileNetV2 inference across two devices with different compute capabilities.
- Modelled the effects of network bandwidth, activation transfer, latency and energy consumption to identify suitable split points under different simulated conditions.
- Compared configurations and found substantial simulated differences between split strategies; results are model-based, not hardware measurements.

Cross-Traffic Management for Autonomous Vehicles 2024 – 2025

UPPAAL, FreeRTOS, ESP32, VHDL | Group project (3)

- Designed a first-come-first-served intersection controller; authored the UML diagrams (requirement, use case, state machine, activity, sequence).
- Built the UPPAAL model of the vehicle–controller handshake (deadlock check passed). Team prototype: FreeRTOS queue on an ESP32 releasing 10 cars in arrival order, and a 16-entry VHDL FIFO simulated in ModelSim.

Education

M.Eng. Embedded Systems Engineering, FH Dortmund, Germany — Sep 2024 – Present

B.Eng. Electronic Engineering, Hochschule Hamm-Lippstadt, Germany — Sep 2019 – Aug 2024

Higher National Diploma, Mechanical Engineering, Yaba College of Technology, Nigeria — 2010 – 2017

Languages

English (native); German (B1, B2 course in progress)